The Hidden Cost of Globalization in the Underwater World: Evidence from California's Kelp Forests

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Abstract

This article examines how a maritime regulation, California's Ocean-Going Vessel Fuel Rule (2009), has affected kelp forests, one of the most important aquatic ecosystems along the California coast. That policy led part of the traffic to move offshore, with potential impact on kelp in the regulated area. Combining satellite measures of kelp canopy and biomass with AIS vessel trajectories, we exploit variation in pre-policy shipping exposure across kelp cells. After the rule, kelp improves disproportionately where maritime traffic was highest. A doubling of pre-policy traffic is associated with about 10–11% more kelp. Event studies show no differential pre-trends, with the treatment effect appearing in 2009, the year of implementation. We apply the same design to the 2011 boundary modification around the Channel Islands, which re-concentrated vessel traffic in specific corridors. Kelp declined most in the cells that had the highest pre-modification shipping exposure. JEL: Q57, R41, F18

Keywords: Kelp forests, maritime regulation, emission control areas, difference-in-differences, ecosystem services, AIS, shipping behavior

1 Introduction

Globalization relies heavily on maritime routes that generate substantial negative externalities and hidden costs. The human health consequences of shipping-related air pollution are now well documented (e.g., Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022), but evidence on impacts on marine ecosystems remains comparatively limited

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at the spatial scale relevant to policy. This paper investigates it by quantifying how a major coastal shipping regulation and the associated reconfiguration of traffic impact the kelp forests in California. Such an analysis contributes to documenting the spatial and environmental footprint of maritime activity.

Kelp forests are large biogenic habitats that support coastal biodiversity and provide multiple ecosystem services (Dayton, 1985). As a foundation species (Dayton, 1985), kelp creates a habitat structure that functions as nursery grounds and supports complex food webs in the world, such as in Europe (Christie et al., 2009) and California (Steneck et al., 2002; Carr and Reed, 2019). As a result, the kelp forest has an economic value through fishery production, nutrient cycling, and carbon sequestration, which, according to Eger et al. (2023), can be estimated at between $465 and $562 billion per year worldwide. Finally, because kelp can respond quickly to changing local conditions through rapid growth and recruitment (Krumhansl et al., 2016; Carr and Reed, 2019), these forests provide us with enough variation over a relatively short horizon to study the impact of vessel traffic changes.

We study California's Ocean-Going Vessel Fuel Rule, which established a regulated zone in July 2009, later modified at the end of 2011. The policy was designed to reduce sulfur-related emissions by requiring the use of low-sulfur fuels within the zone.[1] For our setting, the rule altered vessel routing decisions (Klotz and Berazneva, 2022), thereby changing the spatial footprint of maritime activity near kelp forests. In particular, vessel traffic declines after 2009 and re-concentrates after the 2011 boundary change, which provides quasi-experimental variation to study the sensitivity of kelp forests to maritime exposure.

Our contribution is twofold. First, we combine quarterly satellite-based measures of kelp canopy and biomass with AIS-derived shipping exposure. This continuous difference-in-differences design aims to study whether kelp cells that were more exposed to shipping before 2009 benefit the most from this regulation. We find evidence consistent with this prediction. The coefficient on shipping exposure after the implementation of the policy is positive and significant across all specifications.

Second, we apply the same analysis to the 2011 boundary modification around the Channel Islands, which led traffic to re-concentrate in specific corridors. We find that kelp cells more exposed to vessel traffic before 2011 experienced disproportionately worse outcomes after the boundary change. The two results of this paper are thus consistent. Kelp recovers in the areas that benefited most from the 2009 regulation, but deteriorates in the areas most affected by the 2011 re-concentration of traffic.

This analysis contributes to three different literatures. First, it brings marine forests into the economic analysis of deforestation. A growing literature studies how globalization impacts terrestrial forests (e.g., Berman et al., 2023; Abman and Lundberg, 2020; Carreira et al., 2024; Nedoncelle et al., 2024), but underwater forests have received no attention in economics despite providing comparable ecosystem services (habitat provision, carbon sequestration, and nutrient cycling). By analyzing how maritime traffic

[1] Vessels entering the zone must switch from conventional heavy fuel oil, which can contain up to 3.5% sulfur by mass, to marine distillate fuels capped at 0.1%. The switch must be completed before entering the regulated zone (within 24 nautical miles of the California baseline).

affects kelp forests, we extend the deforestation literature that uses causal inference analysis, to a new biome and a new mechanism. Second, while experimental and site-specific studies have already been done on kelp (e.g., Johansson et al., 2012), a large spatial analysis (along the California coast) related to a maritime policy has never been conducted. Third, while many economic analyses often focus on the impact of pollution on human health (e.g., Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022), we go here beyond this anthropocentric perspective.

2 Background

The California–Baja California coast hosts extensive kelp forests dominated by canopy-forming species (giant kelp, *Macrocystis pyrifera*, and bull kelp, *Nereocystis luetkeana*). These species form surface (or near-surface) canopies that are detectable with satellite imagery (Bell et al., 2023).

Kelp performance is shaped by ocean temperature, nutrient availability, underwater light (or coastal darkening), as well as localized stressors that can damage fronds, inhibit photosynthesis, or alter community structure (Carr and Reed, 2019).

Maritime activity may affect several of these factors, because vessels release pollutants and disturb the water along their routes. Busy corridors may thus be disproportionately hostile to kelp recruitment and growth. In particular, vessel operations generate chemical and nutrient discharges (greywater and blackwater) that can contribute to eutrophication and oxygen depletion (Lindgren et al., 2016), with potential consequences for coastal vegetation (e.g., Filbee-Dexter and Wernberg, 2018).

A second channel operates through the combustion of heavy fuel oil (HFO), which was used before the implementation of California's emission control area (ECA). HFO combustion generates pyrogenic polycyclic aromatic hydrocarbons (PAHs), which are then found in water and accumulate in coastal sediments (Yunker et al., 2002). Appendix A provides spatial evidence for this channel. We find that PAH concentrations are correlated with our AIS-derived measure of pre-policy vessel traffic intensity (Spearman $\rho = 0.55$; Figures 5 and 6).

Shipping can also introduce contaminants through antifouling practices. Biocidal compounds historically used to reduce hull fouling, including tributyltin (TBT) and copper-based formulations, could enter coastal waters and affect marine organisms (Dafforn et al., 2011; Cima and Varello, 2022). Laboratory evidence confirms that antifouling compounds inhibit kelp photosynthesis and growth (Johansson et al., 2012), and field measurements in Southern California harbors document biocide concentrations that may harm macroalgae (Schiff et al., 2011). Finally, maritime traffic can influence nearshore physical conditions (through wake, turbulence and shading).

As a result, we consider that changes in the spatial distribution and intensity of shipping activity can matter for kelp outcomes. California's Ocean-Going Vessel Fuel Rule, which introduced a regulated zone extending 24 nautical miles off the coast, enables us to test this hypothesis. Under this law, vessels must use low-sulfur marine distillate fuels within the zone. Because compliance increases operating costs within the regulated area, this regulation induces changes in the spatial distribution of traffic. More precisely,

using one-minute AIS data for roughly 20,000 voyages per year off the U.S. west coast, Klotz and Berazneva (2022) document a sharp discontinuity at the July 2009 enforcement date. Container ships reduced fuel consumption within the regulated zone by 33–52%, primarily by rerouting outside its boundary and traveling south of the Channel Islands rather than through the Santa Barbara Channel. Finally, in 2011, the southern boundary was modified around the Channel Islands, affecting where ships concentrate as they approach ports and creating additional variation in maritime exposure.

3 The 2009 Ocean-Going Vessel Fuel Rule

This section describes the data and the continuous difference-in-differences design used to estimate changes in kelp canopy and biomass associated with the 2009 implementation of the regulated zone.

3.1 Data

We combine satellite kelp observations, AIS vessel tracking data, and environmental covariates, all aggregated to a common spatial grid of 10 km hexagonal cells at quarterly frequency.

Kelp outcomes and shipping exposure. Kelp outcomes come from Bell et al. (2023), which provides quarterly cloud-free Landsat-based measures of floating kelp canopy at 30 m resolution along the Pacific coast.[2] We use two variables: (i) kelp canopy area within each pixel (m^2) and (ii) kelp canopy biomass (kg) inferred from validated relationships between surface reflectance and in situ canopy and biomass measurements (Bell et al., 2023). Satellite detection captures surface (or near-surface) canopy and does not observe subsurface kelp.

We overlay a uniform grid of 10 km hexagonal cells covering the study area (the 2009 regulated zone and the 2011 boundary modification area) and aggregate all variables to this common spatial unit at the quarterly frequency.[3]

For each year–quarter, the 30 m pixel-level kelp observations are converted to point geometries using their latitude and longitude coordinates and spatially joined to the hexagonal grid. Within each hexagon i and quarter t, we compute the sum and mean of kelp canopy area (m^2) and biomass (kg), as well as the count of kelp pixels.

[2]The underlying imagery consists of Landsat Collection 1 Level-2 Surface Reflectance products, which apply atmospheric correction for aerosols and water vapor *before* kelp classification. Images with more than 25% cloud cover are excluded, and remaining cloud and cloud-shadow pixels are masked using the quality-assurance band. Any localized change in atmospheric transparency, such as reduced sulfur aerosols from the fuel regulation, is therefore corrected at the reflectance stage rather than propagated into the kelp canopy estimates.

[3]Hexagonal grids are preferred over square or rectangular tessellations in spatial analysis because they provide equal area and equidistant centroids, minimize edge effects, and reduce sensitivity to the modifiable areal unit problem (Birch et al., 2007; Openshaw, 1984).

Vessel trajectories are derived from georeferenced ship positions recorded by the Automatic Identification System (AIS), made available by the U.S. Coast Guard.[4] We focus on four major cargo vessel classes (tankers, container ships, general cargo ships, and bulk carriers) and summarize their activity at the hexagon–month level in terms of total distance traveled within the cell ("length", in meters) and number of unique vessel passages ("count"). These two measures capture complementary dimensions of shipping exposure. $Length_{it}$ proxies the cumulative intensity of vessel activity and, by extension, the volume of associated discharges (exhaust emissions, antifouling leachates, greywater), which scale with distance traveled. $Count_{it}$ captures the number of discrete vessel passages, reflecting the frequency of vessel-level events rather than their cumulative intensity. Together, they allow us to distinguish the intensive from the extensive margin of shipping exposure. To assign shipping exposure to kelp locations, we proceed as follows. For each hexagon containing kelp, we create a circular buffer of 20 km radius and identify all shipping hexagons that intersect this buffer. We then aggregate the shipping measures, summing length and count, across all identified shipping hexagons for each kelp hexagon i and quarter t. This yields two continuous exposure variables per kelp cell: total distance traveled by vessels in its vicinity ($Length_{it}$) and total number of vessel passages ($Count_{it}$).

Environmental controls. Kelp forests are affected by multiple interacting environmental drivers, including temperature, water motion, and nutrient availability (Carr and Reed, 2019; Castorani et al., 2022). We include three time-varying physical controls that are well-established determinants of kelp dynamics.

Sea surface temperature (SST) is sourced from MODIS monthly composites and aggregated to quarters. SST is a central determinant of kelp dynamics and is also informative about broader ocean conditions (e.g., Yang et al., 2023; Dunstan et al., 2018).[5] Wave intensity is drawn from ERA5 (Hersbach et al., 2023) and measured as significant wave height (the average height of the highest third of waves), aggregated to quarters; it captures physical stress associated with storms and strong sea states. Precipitation is drawn from ERA5 and aggregated to quarters; it proxies terrestrial nutrient runoff, which can affect kelp growth.

3.2 Specification

Our main design is a continuous difference-in-differences (Callaway et al., 2024) that exploits cross-sectional variation in pre-policy shipping exposure. The intuition is straightforward: kelp cells that were more exposed to vessel traffic before 2009 should benefit more from the implementation of the regulated zone if that regulation effectively reduces nearshore stressors. The treatment intensity is measured by pre-policy shipping activity derived from AIS data: logged mean vessel traffic (measured as total distance traveled)

[4] AIS data accessed via MarineCadastre.gov, jointly maintained by the U.S. BOEM and NOAA.

[5] We do not include a separate upwelling index (e.g., CUTI or BEUTI; Jacox et al., 2018) because SST is strongly negatively correlated with upwelling intensity along the California coast (cold surface water is the primary physical signature of upwelling events) and our year–quarter fixed effects already absorb the dominant seasonal upwelling cycle. Any remaining bias would require a differential change in upwelling at the policy date that is spatially correlated with shipping exposure.

or logged mean vessel count within a 20 km buffer around each kelp hexagon during January–June 2009, before implementation. The regulation was announced before that date, but rerouting ahead of enforcement would have lengthened voyages and raised costs with no offsetting benefit, since the fuel requirement did not apply until July 2009 (Klotz and Berazneva, 2022). The January–June 2009 measure thus records pre-policy routes.

We estimate:

$$Kelp_{it} = \exp(\beta\, Exposure_i \times Post2009_t + \delta\, Z_{it} + f_i + f_t) \cdot \eta_{it}, \tag{1}$$

where η_{it} is a multiplicative error term satisfying $E[\eta_{it} \mid Exposure_i, Z_{it}, f_i, f_t] = 1$, and $Kelp_{it}$ is the sum of kelp biomass (kg) or canopy area (m^2) in hexagonal cell i during quarter t. $Exposure_i$ is the time-invariant, predetermined measure of shipping activity described above (logged mean vessel traffic or logged mean vessel count, January–June 2009). $Post2009_t$ equals one from 2009Q3 onward and zero otherwise. Z_{it} is a vector of log-transformed environmental controls (log sea surface temperature, log precipitation, and log significant wave height).[6] f_i denotes kelp-cell fixed effects (one indicator per hexagonal cell) and f_t denotes year–quarter fixed effects. The cell fixed effects absorb any time-invariant determinant of kelp outcomes, including the main effect of $Exposure_i$, and the time fixed effects absorb shocks common to all cells, including the main effect of $Post2009_t$.

Because kelp outcomes are non-negative and include many zeros, we estimate Eq. (1) by Poisson pseudo-maximum likelihood (PPML; Santos Silva and Tenreyro, 2006).[7] Standard errors are clustered at the cell level (81 clusters).[8] The sample covers Southern California (latitudes 32°–36° N) from 2005 to 2016.[9] The year–quarter fixed effects absorb any coast-wide ecological shock, including the 2013–2016 marine heat wave that caused catastrophic kelp losses in northern California (Rogers-Bennett and Catton, 2019; McPherson et al., 2021). Such a shock would bias β only if its spatial footprint were systematically correlated with the pre-policy shipping-exposure cross-section, for which there is no ecological basis.

The coefficient of interest is β. A positive β estimate would indicate that kelp outcomes improved more strongly after the policy in cells that had been more exposed to vessel traffic beforehand, suggesting that the regulation was most beneficial precisely where maritime activity had been most intense.

[6] All controls are transformed as $\ln(1+x)$. For SST and significant wave height, which are strictly positive and large in magnitude, this is virtually identical to $\ln(x)$. For precipitation, the $\ln(1+x)$ form handles quarters with zero rainfall.

[7] We restrict the sample to cells recording at least two vessel passages per month on average during January–June 2009, excluding cells with negligible shipping exposure. The estimates are unchanged when observations are weighted by log vessel count, and they remain positive and significant when observations are weighted by pre-period kelp biomass (Table 4, Panel D, in Appendix B).

[8] Adjacent cells may share ocean currents and temperature fields, and our exposure measures are built from overlapping 20 km buffers, so errors could be correlated across nearby cells. With Conley spatial-HAC standard errors (Conley, 1999) at cutoffs of 100 km and 200 km, the coefficient remains significant at the 1% level (Table 5 in Appendix B).

[9] Giant kelp (*Macrocystis pyrifera*) dominates the Southern California coast and forms the surface canopy measured by Landsat imagery (Bell et al., 2023). North of approximately 36–37° N, bull kelp (*Nereocystis luetkeana*) progressively replaces it (Dayton, 1985; Steneck et al., 2002).

3.3 Results

Table 1 reports the estimates of Eq. (1) for the two measures of exposure, distance traveled in columns (1)–(2) and vessel count in columns (3)–(4), and for the two kelp outcomes, biomass and canopy area.

	Distance Traveled		Vessel Count	
	(1) Biomass	(2) Area	(3) Biomass	(4) Area
Post 2009 × Exposure	0.1361***	0.1317***	0.1548***	0.1493***
	(0.0359)	(0.0352)	(0.0432)	(0.0424)
SST	-5.703***	-5.512***	-5.728***	-5.537***
	(1.145)	(1.136)	(1.150)	(1.142)
Precipitation	0.0764	0.0812	0.0846	0.0891
	(0.1003)	(0.0981)	(0.1012)	(0.0989)
Significant Wave Height	-3.267**	-3.102**	-3.358**	-3.189**
	(1.367)	(1.344)	(1.367)	(1.342)
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	3880	3880	3880	3880

Notes: Unweighted PPML estimates of Eq. (1). The dependent variable is kelp biomass (kg) or canopy area (m^2). Columns (1)–(2) measure exposure by logged mean distance traveled by vessels within a 20 km buffer around each kelp hexagon during January–June 2009; columns (3)–(4) use the corresponding logged mean vessel count. All specifications include kelp-cell and year–quarter fixed effects. Sample: Southern California (32°–36° N), 2005–2016, restricted to cells with at least two vessel passages per month on average (81 cells × 48 quarters, less 8 cell-quarters with missing covariates in 2015–2016 = 3,880 observations). Standard errors clustered at the cell level (81 clusters) in parentheses. Weighted estimates are reported in Appendix B.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 1: Continuous difference-in-differences: Kelp outcomes and the 2009 ECA

We find that, after the 2009 rule, kelp increased more in the cells that were more exposed to vessel traffic before the policy. The coefficient on exposure interacted with the post-2009 indicator is positive and significant at the 1% level in the four columns, and its size varies little from one column to the other.

In terms of magnitude, a doubling of pre-policy vessel traffic is associated with about 10% more kelp biomass and canopy after 2009 when traffic is measured by distance traveled (columns (1)–(2), $2^{0.136} - 1 \approx 0.10$), and about 11% when it is measured by vessel count (columns (3)–(4)). The two measures describe different aspects of traffic, the cumulative intensity of emissions and discharges for distance traveled and the frequency of passages for vessel count, but they lead to the same conclusion.

Environmental controls enter with expected signs. Sea surface temperature is strongly and negatively associated with kelp outcomes across all columns, consistent with the well-documented sensitivity of kelp to warm-water conditions (Carr and Reed, 2019;

Rogers-Bennett and Catton, 2019). Significant wave height is also negatively associated with kelp, consistent with physical damage to fronds and holdfasts during strong sea states.

The coefficient in Eq. (1) summarizes the response of kelp to exposure with a single linear-in-log term. Following Callaway et al. (2024), we also estimate the average *level* effect of high relative to low exposure, which does not rely on the linear specification. We partition cells into exposure terciles, take the least-exposed tercile as the reference, and estimate group-specific post-2009 effects. We find that, after 2009, kelp in the most-exposed third of cells increased by roughly 46–51% relative to the least-exposed third (Table 6 in Appendix B). The effect is also positive for the middle third.

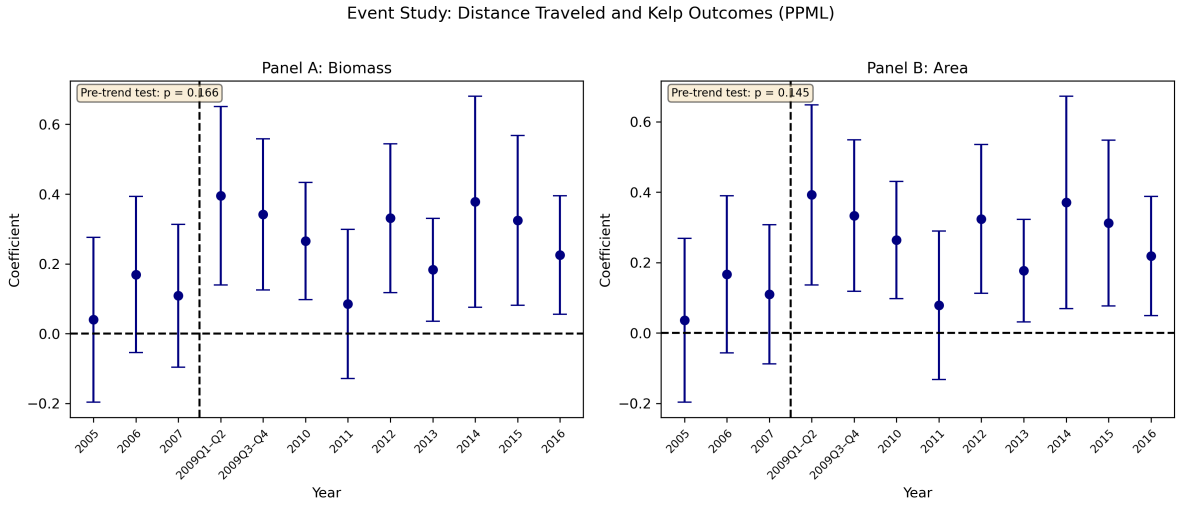
3.4 Dynamic effects

To assess whether the parallel-trends assumption holds and to trace the evolution of treatment effects over time, we estimate an event-study specification:

$$Kelp_{it} = \exp\left(\sum_{\ell \neq 2008} \beta_\ell \, Exposure_i \times \mathbf{1}_{t \in \ell} + \delta \, Z_{it} + f_i + f_t \right) \cdot \eta_{it}, \tag{2}$$

where ℓ indexes event-study periods and $\mathbf{1}_{t \in \ell}$ equals one when quarter t falls in period ℓ. Each calendar year constitutes a single period, except 2009, which is split into Q1–Q2 (pre-enforcement) and Q3–Q4 (post-enforcement). The reference period is 2008. This specification interacts the time-invariant treatment intensity with period indicators, allowing us to trace out the differential evolution of kelp outcomes across cells with different levels of shipping exposure.

Figure 1 reports the estimated coefficients $\hat{\beta}_\ell$ for distance traveled interacted with period indicators, estimated by PPML with the same controls and fixed effects as Eq. (1).



Note: Each panel plots the coefficients from Eq. (2), where distance traveled (logged mean, January–June 2009) is interacted with year indicators, estimated by PPML. The reference year is 2008. The vertical dashed line separates the pre-policy years from 2009, the year of implementation. Error bars are 95% confidence intervals based on standard errors clustered at the cell level. Panel A: kelp biomass; Panel B: kelp canopy area.

Figure 1: Event study: Distance traveled and kelp outcomes (PPML)

Two features of Figure 1 merit attention. First, all pre-policy coefficients (2005, 2006, and 2007) are close to zero and statistically insignificant, and a joint Wald test fails to reject the null of no pre-trend. Second, the coefficients turn positive and significant starting in 2009Q1–Q2, the six months immediately before enforcement, and remain so through the end of the sample (2016), with a temporary dip in 2011. The fact that the effect emerges already in 2009Q1–Q2 is consistent with some operators switching to compliant fuel during the compliance ramp-up period preceding the July 2009 enforcement date. The coefficients are similar in magnitude in 2009Q1–Q2 and 2009Q3–Q4, suggesting a relatively smooth onset rather than a discrete jump at enforcement. A placebo test setting the treatment date at 2007Q3 on the pre-policy sample alone confirms no effect (Table 4, Panel E). The next subsection examines what these patterns imply for identification.

3.5 Threats to identification

Shipping exposure is not randomly assigned across cells. Cells with heavy traffic differ from the others in several respects, and in particular they hold less kelp before the policy (Table 7 in Appendix C). Eq. (1) attributes to the fuel rule any change in kelp that occurs from 2009 onward and varies with exposure. The identifying assumption is therefore that, absent the policy, kelp would have changed in the same proportion in cells with different levels of exposure. The absence of differential trends before 2009 (Section 3.4) supports this assumption, but the pre-policy years cannot be used to test for a factor that operates from 2009 onward. Three factors could vary with exposure over that period:

the recovery of initially depleted kelp stands, the 2008–09 financial crisis, and the 2011 boundary modification. We examine them in turn.

The first factor is the recovery of initially depleted kelp stands. A stand that starts from a low level can grow faster in proportion than a dense one, for purely ecological reasons. This matters here because the specification is exponential, so β measures a proportional effect. Since high-exposure cells hold less kelp initially, such a recovery would produce a positive β without any effect of the fuel rule. Three results indicate that it does not explain our estimates (Table 4, Panel B). First, the correlation between exposure and baseline kelp is moderate (cell-level Pearson $r \approx -0.29$), so the two can be separated in the data. Second, the coefficient remains positive and significant when we control for the initial level of kelp, by interacting log pre-policy biomass with the post indicator, with a linear trend, or with a full set of year effects, or by interacting baseline-kelp quartiles with the post indicator. Third, when the sample is split by baseline kelp, the effect is found in cells with *high* baseline kelp, where there is no depleted stand to recover.

The second factor is the 2008–09 global financial crisis. It reduced trade volumes and port throughput at Southern California ports by roughly 20% in 2009, the year the fuel rule came into force. A fall in vessel calls reduces traffic most in the cells that were most exposed, so part of the kelp gain could come from the fall in trade and not from the regulation. To separate the two, we construct a cell-level measure of port activity from the National Ballast Information Clearinghouse (NBIC) vessel arrival database and add it as a time-varying control. If the fall in trade explained the kelp gain, the exposure coefficient would decrease once port activity is held constant. It changes by at most 2% (Table 4, Panel C), which indicates that the estimated effect does not come from the variation in trade volumes.

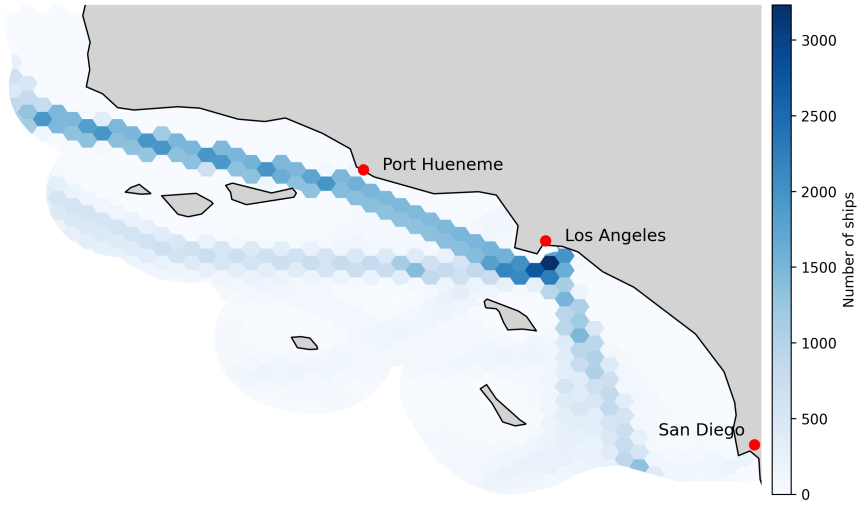
The third factor is the December 2011 boundary modification analyzed in Section 4. Most of the post-2009 window falls after this modification, which changed traffic a second time in part of the same cells, so the post-2009 coefficient could combine the effects of the two policy changes. For this reason, we restrict the sample to 2005–2011, so that the entire post-period precedes the modification. All four coefficients remain positive, significant at the 1% level, and within roughly 10% of their full-sample magnitudes (Table 4, Panel C). This shows that the effect of the fuel rule is already present before the boundary was modified.

4 Boundary Modification of the Regulated Zone in 2011

This section applies the same design to the December 2011 boundary modification around the Channel Islands. The 2009 rule reduced maritime pressure in the cells most exposed to traffic, and kelp improved there. The 2011 modification moved traffic back into specific corridors, so the same relationship predicts that kelp deteriorated in the cells most exposed to that traffic. A result of opposite sign after a policy change of opposite direction is a second test of the relationship between maritime exposure and kelp.

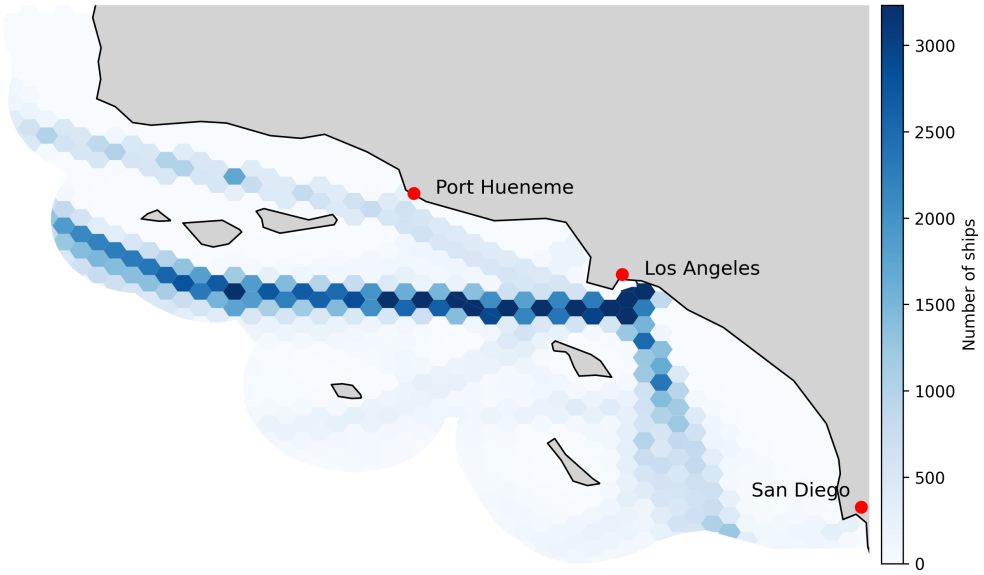
4.1 Spatial redistribution of traffic after 2011

Figures 2 and 3 map vessel traffic in 2009 and in 2011. In 2009, most traffic follows the lanes of the Santa Barbara Channel, along the mainland coast. In 2011, the main flow passes south of the northern Channel Islands, the route that vessels adopted to avoid the regulated zone (Klotz and Berazneva, 2022). The December 2011 modification extended the regulated zone to this southern route.



Notes: Number of vessel passages through each 10 km hexagonal cell in Southern California during 2009. Darker shading indicates higher traffic intensity. Data source: AIS.

Figure 2: Exposure of kelp to maritime traffic in 2009

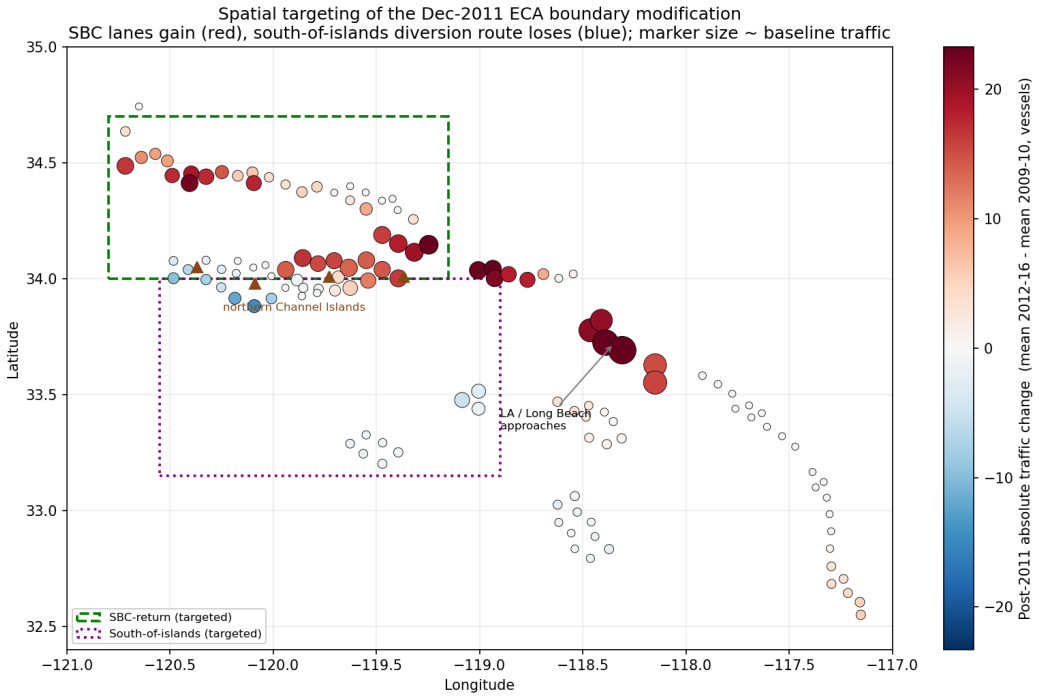


Notes: Number of vessel passages through each 10 km hexagonal cell in Southern California during 2011. The boundary modification took effect in December 2011, so the map overwhelmingly reflects pre-modification routing. Data source: AIS.

Figure 3: Exposure of kelp to maritime traffic in 2011

For the design to be informative, the modification must have increased traffic near the kelp cells that were most exposed to shipping in 2011. We compare mean vessel counts within 20 km of each cell between the pre-modification years (2009–2010) and the post-modification years (2012–2016). Counts rose by roughly twelve vessels in the most-exposed quartile of cells and were essentially unchanged in the least-exposed quartile. In a difference-in-differences on vessel counts with cell and year fixed effects, a doubling of pre-modification exposure is associated with about 2.4 more vessels after the modification.[10] Figure 4 shows where the change occurred. Traffic increased in the Santa Barbara Channel lanes and decreased on the route south of the Channel Islands, whereas a general recovery of shipping would have raised traffic on both routes. We conclude that the modification increased vessel traffic near the kelp cells that were most exposed to shipping in 2011.

[10]The coefficient on log exposure interacted with the post-2011 indicator is 3.48 (standard error 0.56, clustered by cell; 127 cells, 889 cell-year observations, 2009–2016 excluding 2011).



Notes: Each marker is a kelp hexagon centroid; color is the change in mean vessel count within 20 km between the post-modification years (2012–2016) and the pre-modification years (2009–2010), with red indicating gains and blue losses. Marker size is proportional to baseline (pre-modification) traffic. The dashed box outlines the Santa Barbara Channel lanes to which traffic returned; the dotted box outlines the south-of-islands diversion route that the extended boundary brought under regulation. Data source: AIS.

Figure 4: Spatial reallocation of vessel traffic after the 2011 boundary modification

4.2 Empirical strategy

We estimate the same continuous difference-in-differences as in Eq. (1), with exposure measured before the boundary modification. Kelp cells that were more exposed to vessel traffic before the modification should experience worse outcomes afterward if the return of traffic increased nearshore stressors.

We measure shipping exposure over the calendar year 2011, using total distance traveled and vessel count within a 20 km buffer around each kelp hexagon. The boundary modification took effect in December 2011, so eleven of the twelve months in the treatment variable reflect pre-modification routing.[11]

We estimate:

$$Kelp_{it} = \exp\Big(\gamma\, Exposure_i^{2011} \times Post2011_t + \delta\, Z_{it} + f_i + f_t\Big) \cdot \eta_{it}, \tag{3}$$

[11]Measuring exposure in 2010, a year that entirely precedes the modification, gives biomass coefficients of -0.077 (standard error 0.012) for distance and -0.213 (0.046) for vessel count, both significant at the 1% level.

where $Exposure_i^{2011}$ is the log of shipping intensity measured during 2011, $Post2011_t$ equals one from 2012Q1 onward and zero otherwise, and the other terms are defined as in Eq. (1). The equation is estimated by PPML with standard errors clustered at the cell level, on the same area and period as the 2009 specification (latitudes 32°–36° N, 2005–2016). A negative γ indicates that kelp deteriorated more after the modification in cells that were more exposed to shipping before it.

4.3 Results

After the boundary modification, kelp deteriorated more in the cells that were more exposed to shipping in 2011 (Table 2). The table has the same structure as Table 1.

	Distance Traveled		Vessel Count	
	(1) Biomass	(2) Area	(3) Biomass	(4) Area
Post 2011 × Exposure	-0.0925***	-0.0914***	-0.210***	-0.208***
	(0.0288)	(0.0282)	(0.0433)	(0.0425)
SST	-7.155***	-7.060***	-7.103***	-7.008***
	(0.904)	(0.909)	(0.903)	(0.908)
Precipitation	-0.0359	-0.0386	-0.0483	-0.0508
	(0.0781)	(0.0773)	(0.0782)	(0.0776)
Significant Wave Height	-3.452***	-3.371***	-3.341***	-3.262***
	(0.958)	(0.951)	(0.956)	(0.949)
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Observations	6000	6000	6000	6000

Notes: Unweighted PPML estimates of Eq. (3). The dependent variable is kelp biomass (kg) or canopy area (m^2). Columns (1)–(2) measure exposure by logged mean distance traveled by vessels within a 20 km buffer around each kelp hexagon during 2011; columns (3)–(4) use the corresponding logged mean vessel count. All specifications include kelp-cell and year–quarter fixed effects. Standard errors clustered at the cell level (127 clusters) in parentheses. Sample: Southern California (32°–36° N), 2005–2016, 6,000 observations (127 cells). Weighting observations by log vessel count gives biomass coefficients of -0.102 (standard error 0.025) for distance and -0.134 (0.047) for vessel count, both significant at the 1% level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Continuous difference-in-differences: Kelp outcomes and the 2011 boundary modification

The coefficient $\hat{\gamma}$ is negative and significant at the 1% level in the four columns. A doubling of pre-modification vessel traffic is associated with about 6% less kelp biomass and canopy after 2011 when traffic is measured by distance traveled, and about 13.5% less when it is measured by vessel count.

These results have the opposite sign to the 2009 estimates in Table 1. After the 2009 rule, kelp improved more in high-exposure cells ($\hat{\beta} > 0$). After the 2011 modification, kelp deteriorated more in the cells where traffic increased ($\hat{\gamma} < 0$). The two policy changes

moved maritime pressure in opposite directions and kelp responded in opposite directions, which supports a causal interpretation of the relationship between vessel activity and kelp.

Environmental controls enter with expected signs and are consistent with the estimates in Table 1. Sea surface temperature is strongly and negatively associated with kelp outcomes, confirming kelp sensitivity to warm-water conditions. Significant wave height is also strongly negative, reflecting physical stress from storms.

4.4 Dynamic effects

The 2009 fuel rule falls inside the pre-treatment window of the 2011 event, and both designs rely on cross-sectional variation in shipping intensity. We therefore estimate a PPML event study and read it over three periods: a window that precedes either regulation, the 2009 fuel-rule phase-in, and the post-2011 modification period.

We interact 2011 vessel-count exposure with year indicators, with kelp-cell and year–quarter fixed effects and standard errors clustered at the cell level. The reference year is 2011, since the modification took effect in December 2011. Coefficients for 2005–2010 then describe the years before the modification and coefficients for 2012–2016 the years after it, as in Eq. (3).

Table 3 reports the estimates by period. In 2005–2008, before either regulation, the coefficients are individually and jointly insignificant. This indicates that kelp in high- and low-exposure cells followed parallel trends before any policy change. The 2009 and 2010 coefficients are negative and significant. These are the years in which the 2009 fuel rule takes effect, and the cells with high 2011 exposure are largely those that benefited from that rule. After the boundary modification, the coefficients are negative in every year from 2012 to 2016 and significant at conventional levels. Kelp in the cells where traffic increased deteriorated relative to less-exposed cells.

The decline is persistent. The largest coefficients are those of 2014–2016, at the end of the sample, which indicates a lasting change in nearshore conditions once traffic returned to these corridors, as for the 2009 effect.

Panel A: Before both policies					
	2005	2006	2007	2008	Joint test (p)
Biomass	0.090	0.040	0.011	-0.056	0.43
	(0.113)	(0.132)	(0.097)	(0.117)	
Area	0.096	0.049	0.012	-0.049	0.39
	(0.113)	(0.131)	(0.097)	(0.116)	
Panel B: 2009 fuel rule					
	2009	2010			
Biomass	-0.407***	-0.187**			
	(0.088)	(0.091)			
Area	-0.401***	-0.181**			
	(0.089)	(0.091)			
Panel C: After the 2011 modification					
	2012	2013	2014	2015	2016
Biomass	-0.244***	-0.174*	-0.479***	-0.330***	-0.327**
	(0.082)	(0.099)	(0.119)	(0.128)	(0.152)
Area	-0.242***	-0.164*	-0.478***	-0.323**	-0.315**
	(0.081)	(0.099)	(0.120)	(0.128)	(0.152)

Notes: Unweighted PPML estimates from a regression of kelp biomass (kg) or canopy area (m^2) on year indicators interacted with vessel-count exposure (logged mean vessel count within a 20 km buffer, measured in 2011), with kelp-cell and year–quarter fixed effects. The reference year is 2011. The joint test is a Wald test that the four 2005–2008 coefficients are zero. Standard errors clustered at the cell level (127 clusters) in parentheses. Sample: Southern California (32°–36° N), 2005–2016, 6,000 observations.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: PPML event study: dynamics of the 2011 boundary modification

5 Conclusion

> "If in any country a forest was destroyed, I do not believe as many species of animals would perish as would here from the destruction of kelp." — Darwin (1839)

Human reliance on kelp dates back to prehistoric times; Erlandson et al. (2007) even propose a "Kelp Highway Hypothesis" explaining that human migration 20,000 years ago depended on this resource. Yet these same kelp corridors are now traversed by giant vessels that leave maritime pollution in their wake.

In this paper, we have exploited two policy episodes (the 2009 Ocean-Going Vessel Fuel Rule and the 2011 boundary modification around the Channel Islands) to estimate the causal effect of maritime regulation on kelp forests along the California coast, operating through differential shipping exposure. Using a continuous difference-in-differences design with shipping-specific exposure measures derived from AIS vessel trajectories, we show that (i) kelp canopy and biomass improved disproportionately in cells more exposed to vessel traffic after the 2009 regulation tightened fuel standards for nearshore vessels, and (ii) kelp deteriorated disproportionately in cells more exposed to shipping before

the 2011 boundary change re-concentrated traffic in specific corridors. Event-study diagnostics show approximately flat pre-trends in the 2009 specification (Figure 1). For the 2011 episode, a PPML event study dated to the modification confirms parallel pre-trends over the window preceding either regulation and a significant, persistent post-2011 decline (Table 3). The opposite signs of the 2009 and 2011 results (one following a tightening of fuel standards, the other following a spatial reallocation of traffic) strengthen the causal interpretation that maritime activity has a quantitatively important effect on kelp forest health, operating through channels that covary with the spatial footprint of maritime activity, which the regulation altered through both fuel switching and route adjustments.

These findings have implications for both economic analysis and maritime policy. First, as the international trade literature now examines the "hidden costs" of globalization, we bring kelp forests into this discussion: they play an important role in fisheries, alginate production, and carbon sequestration in coastal ecosystems. An illustrative back-of-the-envelope calculation suggests that the canopy gains attributable to differential shipping exposure are worth on the order of $5–10 million per year in ecosystem services, a figure not currently reflected in cost-benefit assessments of maritime emission regulation.[12]

Second, this study supports the objectives of the International Maritime Organization (IMO), which has been pivotal in enforcing ECAs over the past decade. In settings where regulated zones overlap with nearshore ecosystems, ECAs may have effects beyond their impact on air pollution, and could play a role in promoting sustainable international trade that limits damage to coastal habitats. At the same time, the 2011 episode serves as a cautionary finding: boundary modifications that re-concentrate traffic can undo some of the environmental benefits of the original regulation, suggesting that spatial design matters as much as the stringency of emission standards.

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[12] We apply the ecosystem-service valuation of Eger et al. (2023), who estimate that *Macrocystis* kelp forests provide approximately $72,000/ha/year through fishery production, nutrient cycling, and carbon sequestration. We aggregate the estimated effect across cells. For each kelp cell, we recover the canopy attributable to the policy by removing the estimated exposure effect relative to the least-exposed cells, $\text{gain}_i = A_i^{\text{post}}\left[1 - \exp(-\hat{\beta}_A\,(Exposure_i - Exposure_{\min}))\right]$, where $Exposure_i$ is the cell's shipping exposure from Eq. (1), A_i^{post} is observed post-2009 canopy area, and $\hat{\beta}_A$ the canopy-area coefficient. Summing across cells yields roughly 63–135 hectares of attributable canopy, valued at $5–10 million per year. The figure is deliberately conservative: because the year–quarter fixed effects absorb any coast-wide change, it captures only the differential benefit across the exposure gradient and excludes any uniform component of the policy's effect. It also applies a single global per-hectare valuation and treats satellite canopy area as a lower-bound proxy for the service-relevant forest area.

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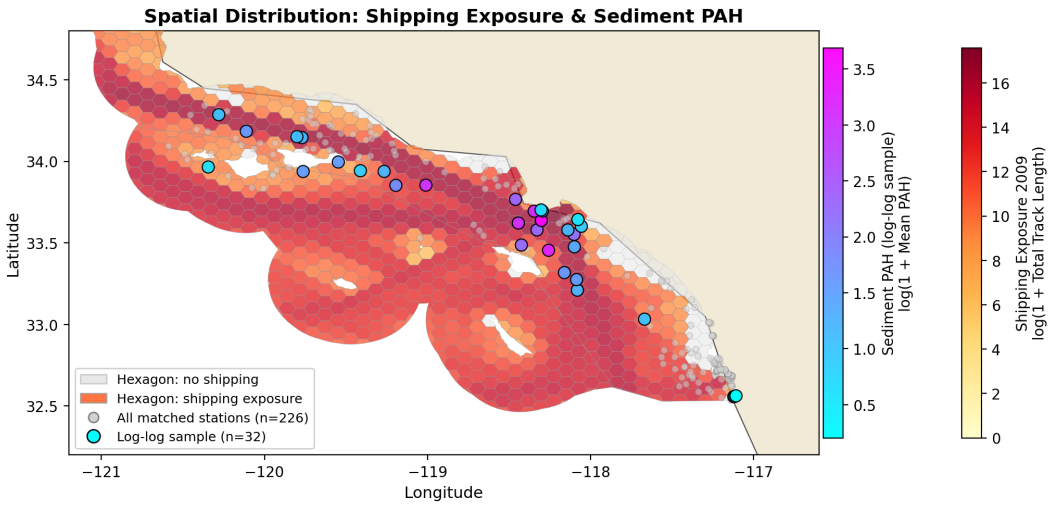
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A Pollution along shipping routes

Section 2 discusses several channels through which maritime activity may affect kelp. The effects of these channels scale with the concentration of vessel traffic. This appendix provides suggestive evidence that our AIS-based treatment variable captures variation in nearshore stressor exposure across kelp cells.

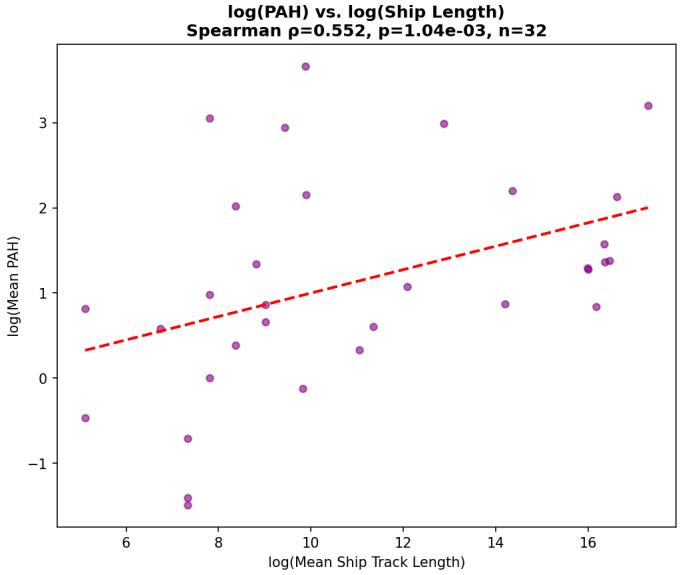
We match our AIS-derived shipping exposure to sediment chemistry stations from the Southern California Bight 2008 Regional Monitoring Program (Schiff et al., 2011), sampled before ECA enforcement. We focus on high molecular weight pyrogenic PAHs, which are produced by petroleum combustion (Yunker et al., 2002).



This figure presents the spatial distribution of shipping exposure and sediment PAH concentrations in the Southern California Bight. Hexagons (ECA grid, 10 km resolution) depict shipping intensity in 2009, measured as total ship track length (log-transformed, YlOrRd color scale). Gray hexagons indicate no recorded shipping activity. Light gray dots represent all sediment monitoring stations (Bight 2008 survey) matched to the nearest shipping grid cell using a nearest-neighbor assignment with a 10 km distance cutoff. Colored dots (cool color scale) highlight the subset of stations with strictly positive PAH concentrations and shipping exposure, used in the log-log regression analysis. Coastline from Natural Earth.

Figure 5: Shipping exposure and sediment PAH concentrations in the Southern California Bight

Stations with the highest pyrogenic PAH concentrations are found in areas of high pre-policy vessel traffic (Figure 5). Across the 32 Bight stations with positive PAH concentrations and shipping exposure, log pyrogenic PAH concentrations are positively correlated with log mean vessel track length (Pearson $r = 0.41$; Spearman $\rho = 0.55$; Figure 6).



Each point represents a sediment monitoring station with strictly positive mean PAH concentration (Bight 2008) and shipping exposure (total ship track length, 2009). Both variables are expressed in natural logarithm. The red dashed line shows the ordinary least squares (OLS) fit. Spearman's rank correlation (ρ), its p-value and the number of stations are displayed in the panel title. Station-to-grid assignment is based on nearest-neighbor matching with a 10 km distance threshold.

Figure 6: Log-log relationship between shipping exposure and sediment PAH concentrations

B Robustness of the 2009 Estimates

Table 4 reports the coefficient on shipping exposure interacted with the post-2009 indicator under alternative specifications of Eq. (1). Panel A reproduces the main estimates of Table 1. Panels B and C correspond to the three factors examined in Section 3.5.

In Panel B, we add controls built from the pre-policy level of kelp to account for the recovery of initially depleted stands: (i) log pre-policy biomass interacted with the post indicator, (ii) baseline-kelp quartiles interacted with the post indicator, (iii) log pre-policy biomass interacted with a linear trend, and (iv) log pre-policy biomass interacted with a full set of year effects. We also estimate the main specification separately on cells above and below the median of pre-policy biomass.

Panel C contains two specifications. The first adds port activity as a control for the 2008–09 financial crisis. Port activity is measured as the log of the inverse-distance-weighted sum of quarterly overseas vessel arrivals at Los Angeles, Long Beach, San Diego and Port Hueneme, from the National Ballast Information Clearinghouse.[13] The second restricts the sample to 2005–2011, the years before the 2011 boundary modification.

In Panel D, observations are weighted by log vessel count or by pre-period kelp biomass (2005–2008 cell average). Panel E assigns a fictitious treatment date of 2007Q3 on the pre-policy sample (2005–2008).

[13] Data available at `https://nbic.si.edu/`.

	Distance Traveled		Vessel Count		
	Biomass	Area	Biomass	Area	Obs.
Panel A: Main specification					
Baseline	0.136***	0.132***	0.155***	0.149***	3880
	(0.036)	(0.035)	(0.043)	(0.042)	
Panel B: Recovery of depleted kelp stands					
+ ln(baseline kelp) × post	0.103***	0.099***	0.114***	0.109***	3880
	(0.038)	(0.037)	(0.043)	(0.042)	
+ baseline-kelp quartile × post	0.122***	0.118***	0.136***	0.130***	3880
	(0.040)	(0.039)	(0.047)	(0.046)	
+ ln(baseline kelp) × trend	0.121***	0.117***	0.135***	0.130***	3880
	(0.036)	(0.035)	(0.042)	(0.041)	
+ ln(baseline kelp) × year effects	0.084**	0.080**	0.095**	0.090**	3880
	(0.037)	(0.036)	(0.042)	(0.041)	
High-baseline cells only	0.129***	0.124***	0.145***	0.139***	1920
	(0.038)	(0.038)	(0.046)	(0.045)	
Low-baseline cells only	0.014	0.012	0.026	0.023	1960
	(0.104)	(0.104)	(0.112)	(0.112)	
Panel C: Financial crisis and 2011 boundary modification					
+ port activity (NBIC arrivals)	0.136***	0.132***	0.152***	0.146***	3880
	(0.036)	(0.035)	(0.043)	(0.043)	
Pre-2011 window (2005–2011)	0.123***	0.122***	0.137***	0.135***	2268
	(0.041)	(0.041)	(0.050)	(0.050)	
Panel D: Weighting					
Weighted by log vessel count	0.135***	0.131***	0.152***	0.146***	3880
	(0.037)	(0.037)	(0.045)	(0.044)	
Weighted by pre-period kelp biomass	0.103***	0.100***	0.124***	0.120***	3736
	(0.030)	(0.029)	(0.038)	(0.036)	
Panel E: Placebo					
Fake treatment date 2007Q3 (2005–2008)	-0.075	-0.069	-0.086	-0.080	1248
	(0.067)	(0.067)	(0.075)	(0.075)	

Notes: Each entry is the PPML coefficient on shipping exposure interacted with the post-2009 indicator in Eq. (1), with the modification indicated in the row label. All specifications include kelp-cell and year–quarter fixed effects and the environmental controls. Baseline kelp is the cell-level mean of pre-policy (2005–2008) biomass. Sample: 81 cells with at least two vessel passages per month on average, 2005–2016 unless indicated; the row weighted by pre-period kelp and the placebo row drop 3 cells with zero pre-period kelp. Standard errors clustered at the cell level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Robustness of the 2009 estimates

Table 5 reports Conley spatial-HAC standard errors, which allow errors to be correlated across cells as a decreasing function of distance. At cutoffs of 100 km and 200 km the exposure coefficient remains significant at the 1% level in every specification.

	Distance Traveled		Vessel Count	
	(1) Biomass	(2) Area	(3) Biomass	(4) Area
Post 2009 × Exposure	0.1361	0.1317	0.1548	0.1493
Cell-clustered SE	(0.0359)***	(0.0352)***	(0.0432)***	(0.0424)***
Conley SE (100 km)	[0.0405]***	[0.0398]***	[0.0478]***	[0.0473]***
Conley SE (200 km)	{0.0296}***	{0.0293}***	{0.0370}***	{0.0369}***
Cell FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	3880	3880	3880	3880

Notes: PPML estimates of Eq. (1) with cell and year–quarter fixed effects. Point estimates are identical to Table 1; the rows report, in turn, the cell-clustered standard error (parentheses) and Conley spatial-HAC standard errors (Bartlett kernel, great-circle distances) at 100 km [brackets] and 200 km {braces} cutoffs. Because a cell's quarterly observations share identical coordinates, within-cell serial dependence is fully absorbed. Sample: 81 cells, 2005–2016.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Spatial-HAC (Conley) standard errors

Table 6 reports group-specific post-2009 effects. We partition the 81 cells into terciles and quartiles of the treatment value, interact each group with the post-2009 indicator, and take the least-exposed group as the reference. The effect is positive at every dosage above the baseline and significant for the top group throughout.

	Distance Traveled		Vessel Count	
	Biomass	Area	Biomass	Area
Panel A: Exposure terciles (base = lowest tercile)				
Middle tercile	0.598***	0.585***	0.275*	0.266
	(0.105)	(0.104)	(0.165)	(0.162)
Top tercile	0.413***	0.400***	0.390***	0.379***
	(0.151)	(0.150)	(0.146)	(0.144)
Panel B: Exposure quartiles (base = lowest quartile)				
Second quartile	0.109	0.110	0.057	0.058
	(0.186)	(0.185)	(0.167)	(0.166)
Third quartile	0.541***	0.522***	0.526***	0.507***
	(0.146)	(0.144)	(0.146)	(0.144)
Top quartile	0.379***	0.367***	0.365**	0.352**
	(0.143)	(0.142)	(0.143)	(0.142)
Linear coefficient	0.136	0.132	0.155	0.149

Notes: Each entry is the PPML coefficient on a dosage-group indicator interacted with the post-2009 indicator, estimated jointly with cell and year–quarter fixed effects and the environmental controls of Eq. (1). Cells are partitioned into terciles (Panel A) or quartiles (Panel B) of the time-invariant treatment value, with the least-exposed group as the omitted reference, so each positive coefficient is the level effect of that dosage group relative to the least-treated cells. The final row reproduces the linear continuous difference-in-differences coefficient from Table 1. Standard errors clustered at the cell level.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Nonparametric dose-response: group-specific post-2009 effects

C Balance Table

Table 7 reports pre-treatment (2005–2008) means of key variables across quartiles of the treatment variable (log vessel distance traveled, January–June 2009). The last column reports the p-value from a Kruskal–Wallis test for differences across quartiles. Mean sea surface temperature and precipitation are close across quartiles, and significant wave height is lowest in the most-exposed quartile. All three are controlled for in the regression. Pre-treatment kelp biomass and canopy area are highest in the least-exposed quartile; Appendix B reports specifications that control for baseline kelp.

	Q1 (low)	Q2	Q3	Q4 (high)	KW p
Log vessel distance (Jan-Jun 2009)	9.311	10.491	12.470	13.321	0.000
	[0.491]	[0.290]	[0.564]	[0.182]	
Kelp biomass (kg)	1655116.8	801416.6	409706.0	321770.5	0.000
	[3976548.5]	[2459511.5]	[929510.4]	[904773.3]	
Kelp canopy area (m^2)	220694.5	107794.3	54520.6	43562.9	0.000
	[521297.4]	[333245.2]	[121129.7]	[123756.5]	
Log SST	2.842	2.883	2.798	2.844	0.000
	[0.134]	[0.124]	[0.125]	[0.124]	
Log precipitation	2.611	2.506	2.779	2.729	0.009
	[1.052]	[1.116]	[1.221]	[1.164]	
Log significant wave height	0.8859	0.7991	0.8732	0.7286	0.000
	[0.1626]	[0.1313]	[0.1748]	[0.1465]	
Log vessel count (Jan-Jun 2009)	0.9722	1.516	3.593	4.300	0.000
	[0.1562]	[0.310]	[0.468]	[0.134]	
Cells	21	20	20	20	

Notes: Cell-quarter means during the pre-treatment period (2005–2008), by quartile of log vessel distance traveled (January–June 2009, 20 km buffer). Standard deviations in brackets. The last column reports the p-value from a Kruskal–Wallis test of equality of distributions across quartiles. Sample: 81 cells (20–21 per quartile), Southern California (32°–36° N), restricted to cells with at least two vessel passages per month on average.

Table 7: Pre-treatment balance across treatment-intensity quartiles